PRATEEK PRADEEP

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PROFILE

Computer Science undergraduate with a strong interest in AI/ML, cloud, and cyber security. Experienced in building and deploying AI-based solutions, including edge/IoT deployment and cloud data synchronization.

EDUCATION

Providence College of Engineering — *2022 – 2026*
Bachelor of Technology in Computer Science — CGPA: 8.51 — *Chengannur, Kerala*

SN Central School — *2020 – 2022*
Senior Secondary School — GPA: 8.7 — *Mavelikara, Kerala*

Delhi Public School — *2018 – 2020*
Secondary School — GPA: 8.74 — *Al Wakrah, Qatar*

PROJECTS

AI-Based Pothole Detection, Alert & Cloud Mapping System

- Trained and optimized a YOLOv8 object detection model using Jupyter Notebook and CUDA acceleration to identify road hazards, achieving high accuracy through data augmentation.
- Deployed the model on a Raspberry Pi 4 IoT edge device using Python and OpenCV, implementing image preprocessing techniques such as CLAHE and green channel filtering to process live video feeds and capture GPS data simultaneously.
- Architected a cloud synchronization system using REST APIs to automatically transmit base64-encoded event images, confidence scores, and JSON metadata to a Firebase Realtime Database.
- Developed a responsive web application to fetch live cloud data, building an interactive UI with dynamic galleries and automated Google Maps integration for pinpointing hazard coordinates.

Voice Control Based Fighting Game

- Developed an accessible 2D fighting game in Unity using C#, integrating the Windows Speech API to allow dual-input combat controls for players with physical disabilities.
- Implemented an event-driven voice recognition system utilizing C# dictionaries to map spoken phrases to in-game actions.
- Developed custom 2D physics and collision detection systems using Rigidbody2D to manage dynamic hitboxes, state-machine animations, and real-time damage calculations.

Linux Hardening & System Security Monitoring

- Developed a network monitoring page as part of a Linux hardening project to enhance system security.
- Built a GUI-based task manager for Linux capable of network monitoring, package management, and displaying CPU/RAM utilization graphs using PyQt5 and matplotlib.

TECHNICAL SKILLS

Programming Languages: C, Java, Python, C#, Dart, Bash/Shell
Frontend: HTML, CSS, JavaScript, React
Backend: Django
AI/ML: YOLOv8, OpenCV, Model Training & Deployment, Jupyter, CUDA
Cloud & IoT: Firebase Realtime Database, REST APIs, Raspberry Pi edge deployment
Database Management: SQL, SQLite, Oracle Database
Developer Tools: Git/GitHub, JupyterLab, WSL, Unity Editor, Linux environment tools

CERTIFICATES

- Python Full Stack with Data Science & Generative AI Certification – Faith InfoTech, Technopark, Trivandrum (Ongoing)
- Presented a research paper and secured acceptance at the ICACA 2026 Conference
- Presented project paper at ICMTIF 2025 Conference
- Google Cybersecurity Professional Certificate — Coursera
- Ethical Hacking on IoT — CTDS India
- Data Structures and Algorithms in Python — NPTEL
- Google Cloud Engineer Learning Path — GDSC
- IEEEXtreme 17.0 and 18.0 (Secured 2nd place in College)
- Bootcamp on 2D Game Development using Unity — TILTEDU